

Dr Sarah Hussain

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Research Profile

Biomedical engineer and postdoctoral researcher specialising in multimodal signal processing, deep learning, and AI-driven systems for clinical and assistive applications. Doctoral research focused on real-time prosthetic control through EMG signal classification, soft tactile sensor fabrication, and embedded AI deployment. Current work spans AI governance in clinical digital health, LLM applications in education, spiking neural network modelling, and wearable health data pipelines. Experienced in designing and executing laboratory experiments, developing end-to-end machine learning pipelines in Python and MATLAB, and translating research into real-world clinical impact. Competitive research funding exceeding £51,000 across EPSRC and Royal Society grants, with publications in IEEE and Frontiers journals and recognition as a STEM for Britain 2026 finalist.

Research Interests

Digital health and AI governance	Myoelectric prosthetic control	Human-machine interaction
LLM applications in clinical and educational settings	Spiking neural networks and neuromorphic computing	Tactile sensing for prosthetics and robotics
Surface EMG and HD-sEMG signal processing	Machine learning for biomedical signals	Rehabilitation robotics and assistive technologies

Education

PhD Medical Engineering 2022 - 2026

Queen Mary University of London | IsDB-QMUL Studentship

Thesis: Multimodal Bio-Signal Processing and Soft Tactile Sensing for Real-Time Prosthetic Control. Viva passed February 2026, minor corrections pending.

MSc Biomedical Engineering 2019 - 2020

Queen Mary University of London | Distinction | UK Commonwealth Scholarship

BEng Biomedical Engineering 2015 - 2018

Mehran University of Engineering and Technology | Distinction

Academic and Professional Appointments

KTP Associate/Postdoctoral Research Associate Oct 2025 - Jul 2026

Queen Mary University of London

- Led and independently completed the full QMUL ethics application, from initial submission through to approval
- Solely responsible for all participant-facing research delivery, including recruitment, screening, informed consent, study visit coordination, and longitudinal follow-up
- Co-developed the patient-facing data collection app with Huma Therapeutics engineering team, identifying clinical and physiological biomarkers based on systematic evidence review
- Identified and specified six digital biomarkers for app integration: validated symptom scores (ACQ, AQLQ), heart rate and HRV, respiratory rate, activity and step count, sleep quality, and environmental triggers
- Developed the analytic validity framework for an LLM-powered asthma coaching chatbot, defining evaluation criteria for clinical accuracy, safety, appropriateness, and response consistency
- Configured and deployed wearable monitoring devices and the Huma patient-facing app, providing participant-facing technical support and liaising with technical partners to resolve device and connectivity issues
- Managed data quality control and electronic data capture within an approved clinical research protocol, maintaining records in compliance with GCP and GDPR
- Secured the AsthmaCoach PPI Studies fund (£1,000) as lead applicant; applied for the Apple Investigator Support Program research kit (decision pending)
- Preparing two first-author journal papers targeting npj Digital Medicine and npj Primary Care Respiratory Medicine

Associate Lecturer*Dec 2025 - Present**Brunel University London*

- Designing and delivering teaching for Advanced Measurement Systems and Systems Modelling and Simulation at MSc level
- Supervising MSc research students in engineering and robotics engineering; 8 students supervised to completion with 5 ongoing
- Setting, marking, and providing feedback on coursework and assessments

Academic Skills Tutor*Feb 2026 - Present**Brunel University London & University of Roehampton*

- Delivering one-to-one and small group academic skills support across Engineering, Computer Science, and Business Schools
- Designing curriculum-aligned workshop materials on critical thinking, academic writing, referencing, and research methods
- Supporting students in developing independent study strategies including time management, exam technique, and study planning

Teaching Fellow, Robotics Engineering*Oct 2024 - Sep 2025**Queen Mary University of London*

- Co-module organiser responsible for 40-55% of delivery across foundation, undergraduate, and postgraduate modules including Robot Design and Mechatronics, Introduction to Robotics, Advanced Robotic Systems, Modelling and Control of Robotic Systems, Cognitive Robotics, Electrical Systems for Renewable Energies, and Science and Engineering Success
- Designed and led hands-on laboratory sessions in MATLAB, Simulink, Python, and OpenCV, covering signal processing, EV systems, and autonomous navigation
- Designed formative and summative assessments with clear rubrics, model answers, and inclusive marking schemes; implemented real-time and written feedback strategies and contributed to moderation processes
- Supervised five MSc dissertation students on interdisciplinary projects in robotics, prosthetics, and biomedical signal processing; two supervisees accepted at CLAWAR 2025 and IEEE MetroXRaine 2025
- Awarded Fellowship of the Higher Education Academy (FHEA) through peer observation and critical reflection; co-authoring a pedagogical paper on generative AI in dissertation supervision (submitted to Computers and Education: AI)
- Acted as personal tutor and academic advisor; conducted regular student check-ins and signposted academic and wellbeing services, advocating for inclusive teaching and students from underrepresented backgrounds

ELRI Network Manager, Early Life Research Institute*Mar 2025 - Aug 2025**Queen Mary University of London*

- Enabled interdisciplinary research development by identifying synergies across life sciences, medicine, public health, and policy
- Supported principal investigators and early-career researchers in preparing and submitting competitive research grants including NIHR, MRC, and UKRI
- Managed institute research communications including newsletters, calls for collaboration, and dissemination of funding opportunities; curated and delivered symposia, sandpits, grant-writing workshops, and research roundtables

Teaching Associate, Medical and Robotics Engineering*Jan 2024 - Aug 2024**Queen Mary University of London*

- Delivered seminars, tutorials, and laboratory sessions across modules including Control System Analysis and Design, Numerical Methods and Data Science in Engineering, Digital Signal Acquisition and Processing, Process Control, Neuromechanics and Bioelectricity of Movement, and Fundamentals of Electrochemical Engineering
- Assessed students, provided feedback, trained demonstrators, and maintained teaching administration

Impact Administration Officer

Sep 2024 - Jan 2025

Queen Mary University of London

- Maintained a comprehensive database of impact grants funded by UKRI, MRC, BBSRC, and HEIF for reporting to respective grant bodies
- Authored impact case study narratives and blogs for successful projects, identifying impactful outputs through quantitative and qualitative data analysis
- Served as panel reviewer for Tech and Engineering impact grants, assessing project outcomes and funding allocations; supported REF-eligible impact case study development

Course Facilitator/Demonstrator

Sep 2022 - Jul 2024

Queen Mary University of London

- Designed and delivered laboratory training sessions across Engineering Design, Experimental Design and Practice, Control System Analysis and Design, Medical Robotics and Surgical Techniques, and Clinical Sensors and Measurements
- Supervised student research projects providing guidance on methodology, timelines, and deliverables

Research Assistant, Better the Bone

Mar 2023 - Jul 2023

Queen Mary University of London

- Co-authored successful EPSRC Impact Acceleration Grant proposal securing £15,000 for biomedical sensing research
- Designed and developed a wearable ultrasound device as a non-invasive alternative to DEXA scanning for osteoporosis diagnosis, targeting pregnant women and patients recovering from fracture
- Designed hardware prototype including ultrasound transducer integration and signal conditioning circuitry
- Developed signal acquisition, processing, and feature extraction pipelines in MATLAB and Python, applying time-frequency analysis and filtering to characterise ultrasound signal propagation through bone tissue
- Validated device performance through controlled laboratory experiments and presented findings at the Francis Crick Innovation Challenge in the Healthy Ageing category

Researcher Development Administrator

Feb 2022 - Dec 2023

Queen Mary University of London

- Managed PhD and postdoctoral training programmes and CPD courses across Queen Mary, coordinating with multiple teams and stakeholders
- Oversaw budget distribution, processed payments, and reported qualitative and quantitative impact to senior management

Publications and Conference Papers

Hussain Khuwaja, S., Dhillon, B.K., Trelfa, J. and De Wilde, J. Generative AI as a Cognitive Partner: Enhancing MSc Engineering Dissertation Supervision through Structured Prompting and Reflective Dialogue. Computers and Education: Artificial Intelligence. [Submitted April 2026]

Hong, C., **Hussain Khuwaja, S.** and Lalitharatne, T.D. Advancements, Challenges and Opportunities in Upper-Limb Prosthetic Sockets: A Comprehensive Review. Assistive Technology. [Under Review]

Usman, S., Shaheed, H. and **Hussain, S.** (2025). Design and Integration of a 15-DOF Myoelectric Prosthetic Hand with Tendon-Driven Actuation and Force Feedback Control. In Climbing and Walking Robots Conference (pp. 73-85). Springer Nature Switzerland.

Zhang, T., Zhou, L. and **Khuwaja, S.H.** (2025). KneeFlex: A Soft Wearable Knee Joint Tele-Rehabilitation Robot. IEEE MetroXRINE 2025 (pp. 776-781). IEEE.

Khuwaja, S.H., Dhillon, B., Mughal, M.M., Hussain, S. and Shaheed, M.H. (2025). Towards Real-Time Prosthetic Control Using Multi-Domain Time-Frequency EMG Classification and ResNet Transfer Learning. IEEE MetroXRINE 2025 (pp. 889-894). IEEE.

Batool, R., Ali, S., Mahnoor, M., Chandio, R., Ursani, A.A., **Hussain, S.** and Memon, A.A (2024). Monitoring Carbon Dioxide Levels in Exhaled Breath for Biomedical Applications. In Proceedings of the 4th International Conference on Key Enabling Technologies (KEYTECH 2024) (p. 92).

De Wilde, P., Forasacco, E., Freire, J., **Hussain, S.** and De Wilde, J. (2023). Generative AI, ChatGPT, and the Impact on Postgraduate Work. Panel discussion, UKCGE, 21 April 2023.

Grants and Funding

Royal Society International Exchange Grant: Flexible Ferroelectret Metamaterials for Healthcare Robotics with Tsinghua University	£12,000 2025
EPSRC Exploratory Grant: Soft Robotic Inspired Sockets for Upper-Limb Prosthetics	£25,761 2025
Asthma Coach Investigator Support Program: Equipment in-kind	£36,372 2026
Centre for Public Engagement Grant: EMBODY	£1,000 2025
Centre for Public Engagement Grant: Asthma Coach PPI Studies	£1,000 2026
EPSRC Impact Acceleration Grant: Better the Bone	£12,000 2023

Technical Skills

- Programming: Python, MATLAB, C, Arduino, SQL (working knowledge)
- Deep learning: TensorFlow, Keras, PyTorch - CNNs, LSTMs, ResNet, ViT, transfer learning, TFLite embedded deployment
- Signal processing: CWT, STFT, time-frequency analysis, band-pass and notch filtering, z-score normalisation, FFT spectral analysis, ultrasound signal processing
- Computational neuroscience: Izhikevich spiking neuron model, neuromorphic encoding, SimScape, Simulink, biomechanical modelling
- Data analysis: NumPy, SciPy, PyWavelets, ablation studies, cross-validation, statistical analysis
- Embedded AI: Teensy 4.1, Arduino, Raspberry Pi, real-time inference, TinyML, sub-millisecond latency deployment
- Sensor integration: sEMG, HD-sEMG, IMU/MEMS, piezoelectric textile sensors, ultrasound transducers, force-sensitive resistors
- Clinical research: GCP, GDPR, IRAS, ethics applications, data quality control, regulatory compliance
- Cloud and tools: Google Cloud (working knowledge), Git, version control, reproducible pipeline development
- Design tools: CREO, CAD, 3D printing, soft sensor fabrication using PDMS and Ecoflex elastomers

Research Talks and Poster Presentations

- STEM for Britain, Houses of Parliament (2026) Selected finalist, presented AR prosthetics research to Members of Parliament
- Russell Binions PhD Research Symposium, QMUL (2025) SEMS PhD Research Excellence Award / Best Presentation Prize (1st Prize)
- EPSRC Bionics+ Sandpit, University of Warwick (2024) Invitation-only competitive sandpit; contributed to SOFTSocket grant
- DERI Sandpit: Sensation and Perception in Biological and Artificial Intelligence, QMUL (2024)
- 60 Years of Robotics at QMUL Symposium (2024) Research talk on myoelectric prosthetic systems
- Hamlyn Symposium on Medical Robotics, Imperial College London (2024) Poster presentation
- 5th UK Manipulation Workshop, University of Oxford (2024) Poster presentation
- BioMedEng24 Conference, QMUL (2024) Three poster presentations including IMU gait analysis and HD-sEMG
- IEEE UK & Ireland RAS Chapter Conference (2023) Supervised and mentored student presenter

Student Supervision

MSc Dissertation Supervision (13 students):

- Reinforcement learning based locomotion control for wireless capsule endoscopy

- Multimodal haptic companion systems for surgical training
- SoftHap multimodal haptic-integrated soft robotic prosthetic hand
- Computer Vision and Augmented Reality for Real-Time Feedback in Laparoscopic Skills Training
- AI Engine for Multimodal Analysis of Training, Nutrition and Recovery Content (completed)
- Design and Development of an AI-Integrated Myoelectric Prosthetic Hand with 27 DoF (completed)
- KneeFlex: A Soft Wearable Knee Joint Tele-Rehabilitation Robot (completed, published IEEE MetroXRaine 2025)
- Integrating Emotion Recognition and Gesture Interaction for Adaptive Rehabilitation Gaming (completed)

PhD Co-supervision

- Tactile sensing and neuromorphic encoding for sensory feedback in prosthetic hands

Awards and Recognition

- STEM for Britain Finalist, Houses of Parliament (2026)
- Francis Crick Innovation Challenge Award: Healthy Ageing (Better the Bone project)
- SEMS PhD Research Excellence Award / Best Presentation Prize, Russell Binions Symposium (1st Prize, 2025)
- Commonwealth Scholarship

Teaching Qualifications

- Fellow of the Higher Education Academy (FHEA) Advance HE, 2025
- Associate Fellow of the Higher Education Academy (AFHEA) Advance HE, 2024

Professional Development and Training

- Good Clinical Practice (GCP), NIHR, 2025
- CN3 Computational Neuroscience, Neurotechnology and Neuro-inspired AI Summer School, Ulster University, 2025
- Vitae International Researcher Development Conference, 2024
- EPSRC Bionics+ Sandpit, University of Warwick, 2024
- Spatial Data Science across Languages Workshop, University of Munster, Germany, 2023
- QIncubator Entrepreneurship Training, QMUL, 2026 (shortlisted)

Professional Memberships

- IEEE Member, Engineering in Medicine and Biology Society; Robotics and Automation Society; IEEE Young Professionals
- Institute of Engineering and Technology (IET) Member

References

Available upon request